

Heat balance in Titan's atmosphere

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Abstract

The recent measurements of the vertical distribution and optical properties of haze aerosols as well as of the absorption coefficients for methane at long paths and cold temperatures by the Huygens entry probe of Titan permit the computation of the solar heating rate on Titan with greater certainty than heretofore. We use the **haze model** derived from the **Descent Imager/Spectral Radiometer (DISR)** instrument on the Huygens probe [Tomasko, M.G., Doose, L., Engel, S., Dafoe, L.E., West, R., Lemmon, M., Karkoschka, E., See, C., 2008a. A model of Titan's aerosols based on measurements made inside the atmosphere. *Planet. Space Sci.*, this issue, doi:10.1016/j.pss.2007.11.019] to evaluate the variation in **solar heating rate** with altitude and solar zenith angle in Titan's atmosphere. We find the disk-averaged solar energy deposition profile to be in remarkably good agreement with earlier estimates using very different aerosol distributions and optical properties. We also evaluated the **radiative cooling rate** using measurements of the thermal emission spectrum by the **Cassini Composite Infrared Spectrometer (CIRS)** around the latitude of the **Huygens site**. The thermal flux was calculated as a function of altitude using temperature, gas, and haze profiles derived from Huygens and Cassini/CIRS data. We find that the cooling rate profile is in good agreement with the solar heating profile averaged over the planet **if the haze structure is assumed the same at all latitudes**. We also computed the solar energy deposition profile at the 10°S latitude of the probe-landing site averaged over one Titan day. We find that some **80% of the sunlight that strikes the top of the atmosphere at this latitude is absorbed in all**, with 60% of the incident solar energy absorbed below 150 km, 40% below 80 km, and 11% at the surface at the time of the **Huygens landing near the beginning of summer in the southern hemisphere**. We compare the radiative cooling rate with the solar heating rate near the Huygens landing site averaging over all longitudes. At this location, we find that the solar heating rate exceeds the radiative cooling rate by a maximum of 0.5 K/Titan day near 120 km altitude and decreases strongly above and below this altitude. Since there is no evidence that the temperature structure at this latitude is changing, the general circulation must redistribute this heat to higher latitudes.

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1. Introduction

The forcing for atmospheric dynamics in planetary atmospheres is due to the difference between the solar heating rate and the thermal cooling rate and their variations with altitude, longitude, and latitude. **The solar heating rate in Titan's stratosphere and below** is due to **absorption by haze aerosols** at visible wavelengths and **by methane** at longer wavelengths. Both of these quantities have recently been measured with greater accuracy than

has been available in the past. The **Descent Imager/Spectral Radiometer (DISR)** instrument on the **Huygens entry probe of Titan measured the upward and downward flux of sunlight at wavelengths between 350 and 1600 nm for altitudes from 150 km to the surface**. These measurements have been combined with measurements of the solar aureole and the polarization near 90° scattering angle to constrain the vertical distribution, single-scattering albedo, and **size and shape of the fractal aggregates that comprise the thick haze on Titan (Tomasko et al., 2008a)**. Also, the optical measurements on the probe provided improved values for the absorption coefficients of methane at the long paths and cold temperatures of Titan (Tomasko et al.,

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2008b). Thus, even though the measurements on the orbiter have yet to be interpreted in the light of the probe measurements to determine the variations in the haze (and possibly methane-mixing profile) over the disk of Titan, the new probe measurements offer the opportunity to evaluate the solar heating rate and its variations with altitude for comparison with past estimates and as more certain inputs to dynamical models.

Here we evaluate the solar heating rate using the DISR haze model and the new methane absorption coefficients as a function of the solar zenith angle. Assuming that the haze has the same properties everywhere on the disk, we compute the average net flux profile and the average solar heating profile for comparison with past estimates based on very different aerosol properties and distributions.

In addition to the solar heating rate, the Cassini mission offers the opportunity to determine the thermal radiative cooling rate. This rate depends on the temperature profile, the composition, and the opacity of aerosols at thermal wavelengths. This information has become available from Titan observations using the Composite Infrared Spectrometer (CIRS) on the orbiter. We compute the radiative cooling rate from an atmospheric model based on Huygens measurements and CIRS observations around 10–15°S and compare this cooling profile with the disk-averaged solar heating profile.

Finally, we compute the solar heating profile averaged over one Titan day at the probe-landing site and compare it with the thermal cooling profile at the same location. As improved information on the variations of distribution of haze, temperature structure, and composition become available from continuing orbiter observations, extensions of these computations will improve our understanding of atmospheric dynamics, heat balance, and haze production and transport over the disk of Titan.

2. Computation of solar heating rate profile

We use the haze distribution and optical properties of Tomasko et al. (2008a, b) to compute the net solar flux profile as a function of wavelength, altitude, and solar zenith angle. Let $I(z, \mu, \phi, \lambda)$ be the intensity of sunlight at altitude z km above the surface of Titan heading downward from a zenith angle θ whose cosine is μ , from an azimuth angle ϕ from the azimuth of the sun, at a wavelength λ microns. The units of intensity are $\text{W}/(\text{m}^2 \text{sr} \mu\text{m})$. Then the downward diffuse flux of sunlight is given by

$$\mathcal{F}_{\text{Down Diffuse}}(z) = \int_{300 \text{ nm}}^{3000 \text{ nm}} \int_0^{2\pi} \int_0^1 I(z, \mu, \phi, \lambda) \mu d\mu d\phi d\lambda. \quad (1)$$

The downward direct flux of sunlight is given by

$$\mathcal{F}_{\text{Down Direct}}(z) = \int_{300 \text{ nm}}^{3000 \text{ nm}} \mu_0 \mathcal{F}_0 \exp^{-\tau(\lambda)/\mu_0} d\lambda. \quad (2)$$

The upward diffuse flux of sunlight is given by

$$\mathcal{F}_{\text{Upward Diffuse}}(z) = \int_{300 \text{ nm}}^{3000 \text{ nm}} \int_0^{2\pi} \int_{-1}^0 I(z, \mu, \phi, \lambda) \mu d\mu d\phi d\lambda. \quad (3)$$

The total downward flux of sunlight is given by the sum of Eqs. (1) and (2). Since for sunlight the total downward flux is greater than the total upward flux, we define the net flux of sunlight to be directed downward and to be given by

$$\mathcal{F}_{\text{net solar flux}}(z) = \mathcal{F}_{\text{Down Diffuse}}(z) + \mathcal{F}_{\text{Down Direct}}(z) - \mathcal{F}_{\text{Upward Diffuse}}(z). \quad (4)$$

The computations from 300 to 1600 nm are done with a plane-parallel layer doubling and adding code using the new methane coefficients. Comparisons with spherical shell models indicate that the errors due to the extended spherical nature of Titan's atmosphere are relatively minor below altitudes of about 200 km where the DISR aerosol models are appropriate. At higher altitudes, corrections for the spherical nature of the atmosphere will be required. We extend the computations to beyond 3000 nm to include the relatively small amount of sunlight at these longer wavelengths. Beyond 1600 nm, we use methane absorption coefficients from Irwin et al. (2006).

The net solar flux at any altitude is the solar energy in W/m^2 that is absorbed below that altitude, and the difference in the net solar flux at two altitudes divided by the thickness of the layer gives the solar energy absorbed in the intervening layer in W/m^3 . In Fig. 1 we show the solar energy deposition rate (before the integration over

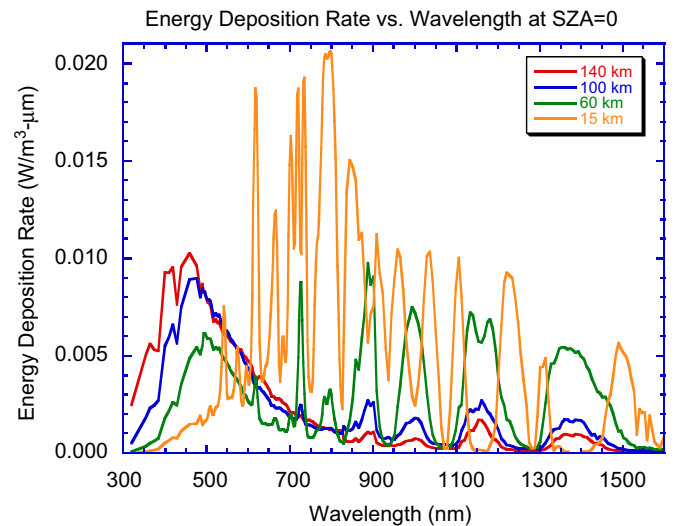


Fig. 1. The solar energy deposition rate (in $\text{W}/\text{m}^3\text{-}\mu\text{m}$) as a function of wavelength at several altitudes is shown for several altitudes at a solar zenith angle of zero. Note that at high altitudes the heating comes primarily from wavelengths short of 700 nm. The energy deposition rate decreases as the altitude decreases in this wavelength range. At longer wavelengths the strong methane bands provide strong heating from 100 to 60 km before decreasing to the surface. The weak methane bands provide the bulk of the heating in the troposphere.

wavelength) with the sun at the zenith at a few altitudes to show the wavelength dependence of the absorbed solar energy. Here the sunlight deposited in $\text{W/m}^3\text{-}\mu\text{m}$ shows that a large fraction of the absorbed solar energy comes from the visible and blue wavelengths at altitudes from 80 to 140 km where the aerosols are most absorbing and the cumulative methane abundance is small. At lower altitudes, the stronger and then the weaker regions of methane absorption contribute larger fractions of the absorbed solar energy. In the strongest methane bands, little or no sunlight penetrates to the surface even with the sun overhead, and the heating at low altitudes in the strongest methane bands is small, while the weaker shoulders of the strong bands and the weaker methane bands provide the largest fraction of the absorbed solar energy.

The net solar flux profiles can be integrated over wavelength (as in Eqs. (1)–(4)) to give the total solar net flux at each altitude over all wavelengths as a function of the zenith angle of the sun and then the solar energy deposition rate. Fig. 2 shows such solar energy deposition profiles for different values of the solar zenith angle. Note that the relative proportion of solar energy deposited at lower altitudes increases as the solar zenith angle decreases. The structure in these curves results from the details of the aerosol vertical distribution.

Even though we well know that the haze structure varies somewhat between the northern and southern hemispheres and especially at high latitudes, it is interesting to integrate the Huygens haze model over the disk for comparison with older models that used very different haze vertical distributions and optical properties. Integrating the net solar flux from the DISR haze model over the disk and dividing by the total surface area of the planet gives the disk-averaged net solar flux profile shown in Fig. 3.

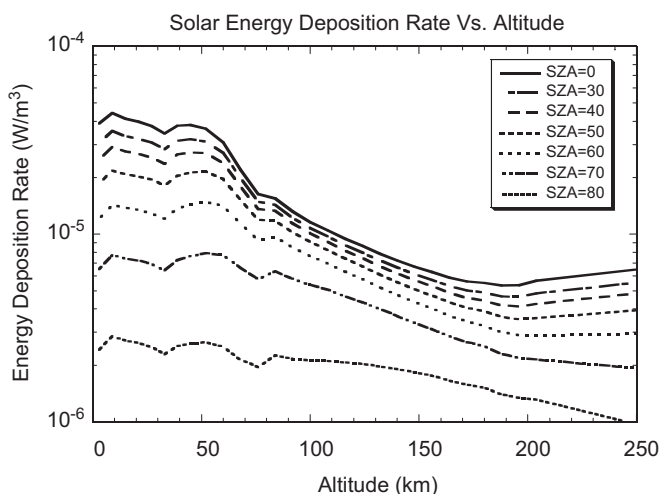


Fig. 2. The wavelength-integrated energy deposition rate profile is shown as a function of altitude for several solar zenith angles, as labeled. The solar energy deposition rate peaks at low altitudes for small solar zenith angles and this effect decreases with increasing solar zenith angle. Some structure is seen in the profiles due to the structure in the haze distribution and the increasing amount of methane below 30 km.

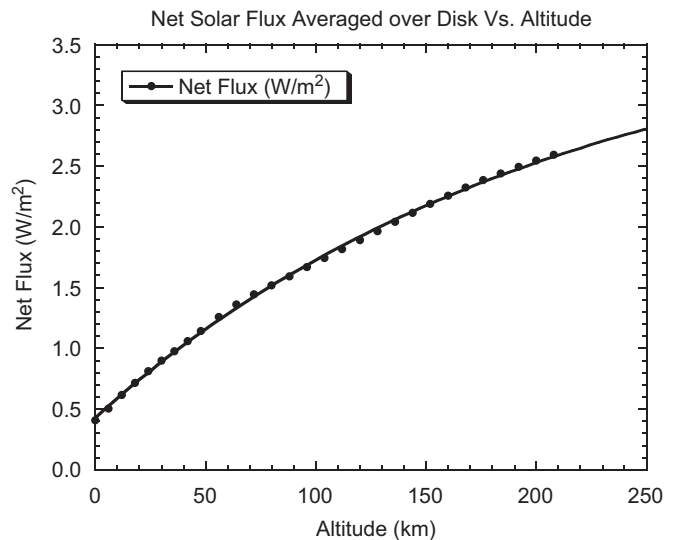


Fig. 3. The net solar flux is shown averaged over the entire disk of Titan (including the dark portion of the disk) as a function of altitude.

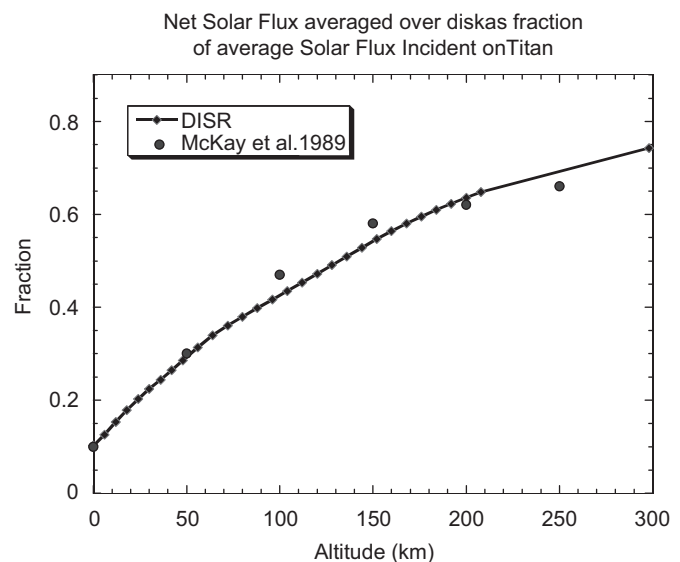


Fig. 4. The net solar flux averaged over the entire disk of Titan as a fraction of the average solar flux incident on the top of Titan's atmosphere is shown as a function of altitude. The average solar flux at the top of the atmosphere is evaluated as πR^2 times the solar flux at the top of the atmosphere divided by the surface area of Titan, $4\pi R^2$. The curve is the fraction computed from the DISR haze model. The points are from McKay et al. (1989). The two profiles are in reasonably good agreement.

This profile can be compared with earlier estimates. Fig. 4 shows the “average” net solar flux profile from McKay et al. (1989) as a fraction of the average sunlight ($\pi R^2 \mathcal{F} / 4\pi R^2$, where $\mathcal{F}$ is the solar flux at the top of Titan's atmosphere and R is the radius of Titan) incident on Titan. Note that the McKay et al. estimate is in remarkably good agreement with the disk-averaged net solar flux profile divided by the average solar flux at the top of the atmosphere ($\mathcal{F}/4$) that results from the new aerosol

model. The profile shows that some 10% of the solar energy that is incident on the disk of Titan is absorbed at the surface, while about an additional 40% is absorbed between the surface and 130 km altitude. An additional roughly 25% of the solar energy that strikes Titan is absorbed at higher altitudes.

The solar heating rate can be given by the energy deposited per unit volume divided by the heat capacity of the gas per unit volume of the gas. Since the gas density increases by many orders of magnitude from the stratosphere to the surface, the heating rate will decrease accordingly with decreasing altitude if only a bit more solar energy is deposited at low compared to high altitudes. Thus, the heating rate decreases very rapidly with decreasing altitudes. This is shown in Fig. 5 for various values of the solar zenith angle. The same percentage changes in the heating rate occur due to changes in aerosol structure as in Fig. 2 for the solar energy deposition profile, but the large variation in heat capacity with altitude requires many decades to display the heating rate curves, and the structure in the heating rate profile seems less in this display than in Fig. 2.

Fig. 6 shows the solar heating rate profile (in K/Titan day) averaged over the disk. Also shown here is the net thermal cooling rate profile computed near the latitude of the Huygens probe-landing site described in the next section. Note that these two profiles are remarkably similar.

3. The thermal radiative cooling rate profile

Calculation of the cooling rate profile requires a precise knowledge of the temperature profile, the vertical profiles

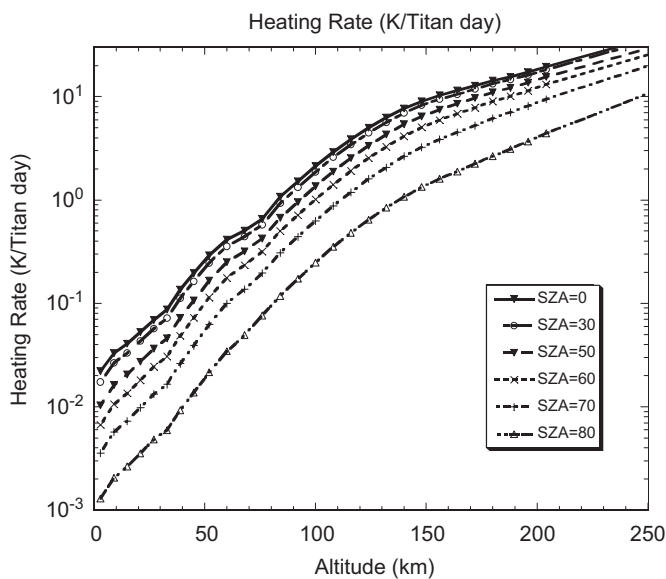


Fig. 5. The calculations of solar energy deposition rate as a function of altitude in Titan's atmosphere are divided by the heat capacity in each layer to give the profiles of solar heating rate (in units of K/Titan day) as functions of solar zenith angle.

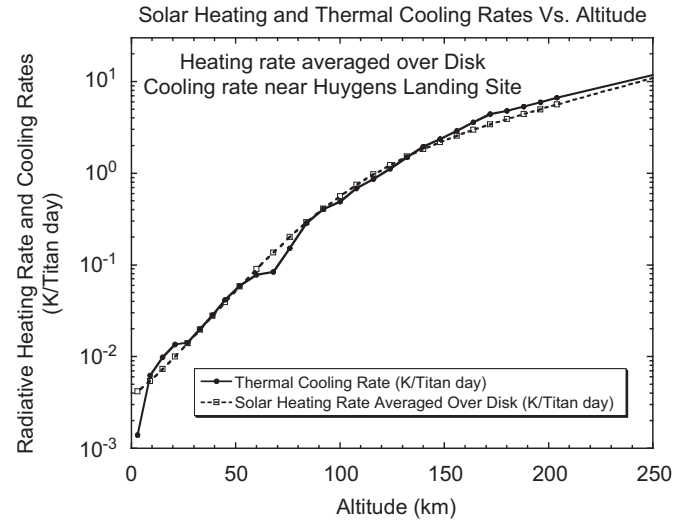


Fig. 6. The profiles of net flux used to make Fig. 1 are averaged over the disk of Titan, differentiated, and divided by the heat capacity in each layer to give the solar heating profile averaged over the disk of Titan (dashed curve). The thermal cooling rate computed near the Huygens landing site is shown (solid curve). Note that the two curves are in remarkably good agreement except where the thermal cooling curve is most uncertain between 70 and 100 km and near the surface as described in the next section. At 200 km and above the solar heating curves rely on uncertain models of the distribution of haze aerosols above the 140 km altitude where Huygens measurements began.

of all significant infrared emitters (gases and aerosols), and the associated spectroscopic parameters. To do so, we used information provided by the Huygens probe complemented with spectral observations of Titan's thermal emission by Cassini/CIRS. The spectral selections for CIRS Focal Planes FP3 (600–1100 cm^{-1}) and FP4 (1100–1400 cm^{-1}) are those described in Vinatier et al. (2007a). The observations were conducted at high-spectral resolution (0.52 cm^{-1}) during the Tb flyby in a latitude range 10–15°, both in nadir and limb-viewing geometries. In addition, we used selections of FP1 CIRS spectra (10–600 cm^{-1}) to cover the long-wavelength range of Titan's thermal emission. To ensure high enough S/N ratios, we selected data recorded at low-spectral resolution (15.6 cm^{-1}) and averaged a large number of them. From spectra recorded during the T6 flyby between 5 and 15°S, we made two selections. A first sample consists of 578 spectra recorded at low emission angles, with a mean of 14°, while the second one incorporates 107 individual spectra at a mean emission angle of 59.5°.

The cooling rate profile was calculated from the divergence of the net thermal flux $d(\pi F)/dp$. Neglecting scattering, the thermal flux at pressure level p is given by

$$\pi F = 2\pi \int_0^\infty d\sigma \left[\int_{\tau_\sigma}^\infty B_\sigma(T') E_2(\tau'_\sigma - \tau_\sigma) d\tau'_\sigma - \int_0^{\tau_\sigma} B_\sigma(T') E_2(\tau_\sigma - \tau'_\sigma) d\tau'_\sigma \right], \quad (5)$$

where τ_σ is optical depth at wavenumber σ and pressure level p , $B_\sigma(T')$ is Planck function at wavenumber σ and temperature T' of pressure level p' , and E_2 is the exponential integral of second order. Note that the net thermal flux is defined as the upward thermal flux minus the downward thermal flux, and is generally positive throughout the atmosphere. Recall that our convention for the net solar flux is opposite to this, and is equal to the larger total downward solar flux minus the upward solar flux (from Eq. (4)). The net thermal flux was calculated from 10 to 1400 cm^{-1} by means of a line-by-line radiative transfer model, with LTE assumed in the whole atmospheric grid (0–540 km). We included collision-induced absorption from various combinations of N_2 , CH_4 , and H_2 pairs, opacity from rovibrational bands of CH_4 , CH_3D , C_2H_6 , C_2H_2 , C_2H_4 , C_3H_8 , $\text{CH}_3\text{C}_2\text{H}$, C_4H_2 , HCN , and CO_2 , rotational lines of CH_4 , HCN , and CO , and opacity from haze particles, assumed to be pure absorbers. References for the spectroscopic parameters can be found in Vinatier et al. (2007a).

We used the temperature profile retrieved by Vinatier et al. (2007b) by means of an inversion algorithm applied to FP4 CIRS nadir and limb spectra of the ν_4 methane band. A uniform stratospheric mole fraction for methane of 1.4% was used in the retrieval process (Niemann et al., 2005). These spectra provide information in the altitude range 120–500 km. At lower altitudes, the retrieved profile smoothly merges into the HASI temperature profile (Fulchignoni et al., 2005). The surface temperature was taken as 93.5 K, equal to that measured in the air after landing.

The CH_4 profile in our atmospheric model is that measured in situ by the Huygens/GCMS (Niemann et al., 2005). The CH_3D profile corresponds to a constant D/H ratio of 1.32×10^{-4} (Bézar et al., 2007). The CO mole fraction, constant with height, is 47 ppm (de Kok et al., 2007a). The vertical abundance profiles of the other absorbers listed above were retrieved from the CIRS FP3 spectra ($600\text{--}1100\text{ cm}^{-1}$) following the procedure of Vinatier et al. (2007a). Information on these constituents is available between approximately 100 and 460 km, i.e. down to about their condensation levels. Continuum emission from the stratospheric haze is clearly visible in the FP3 and FP4 spectra both in nadir and limb-viewing geometry. DISR data indicate the haze optical depth decreases above 80 km with a scale height $H = 65$ km. We kept the same vertical dependence to model the CIRS spectra. We checked that the continuum emission in the limb spectra is adequately reproduced with an optical depth profile having $H = 65$ km. A detailed analysis of the limb spectra in terms of haze opacity is deferred to future publication. The total optical depth above 80 km was set to reproduce the continuum level in the nadir FP3 and FP4 spectra at selected wavenumbers in the interval $600\text{--}1400\text{ cm}^{-1}$ (outside of the emission bands) and linearly interpolated between them. Below 80 km, the DISR dataset shows that the spectral characteristics and vertical distribution of the

particles change. In the lack of information from CIRS in this region, due to the prevailing low temperatures, we arbitrarily assumed that the optical depth then increases linearly with depth, i.e. a constant particle number density was used. Tests were made in which this particle number density was changed by a factor of 3 (see below).

To model the thermal flux in the long-wavelength range of CIRS (FP1: $10\text{--}600\text{ cm}^{-1}$), we considered the collision-induced absorption from $\text{N}_2\text{--CH}_4\text{--H}_2$ pairs, which is the dominant source of opacity in the troposphere and stratospheric haze emission. The use of two spectral selections at different emission angles allows us, to some extent, to disentangle the two contributions. A uniform H_2 mole fraction of 0.1% was used, a value that best reproduces the broad $S(0)$ transition at 350 cm^{-1} in the CIRS spectra and agrees with the Voyager-derived values (Courtin et al., 1995; Samuelson et al., 1997). We found that the $150\text{--}450\text{ cm}^{-1}$ region, which is dominated by $\text{N}_2\text{--CH}_4$ absorption, cannot be reproduced with the Huygens/GCMS methane profile if the absorption coefficients provided by Borysow (<http://www.astro.ku.dk/~aborysow/programs/>, Borysow and Tang, 1993) are used. We then chose to multiply this absorption by a factor of 1.50 to restore a satisfactory fit. The same problem was noted by Courtin et al. (1995) and Samuelson et al. (1997) who then concluded the presence of a large supersaturation of methane in the troposphere. As it was not observed by the Huygens GCMS, we believe instead that Borysow's $\text{N}_2\text{--CH}_4$ coefficients are probably in error, possibly due to the lack of laboratory measurements below 125 K. As concerns stratospheric haze emission, we proceeded as for our FP3 and FP4 modeling and fitted the continuum level in our two selections at 100, 200, 300, 400, 500, and 600 cm^{-1} using a haze scale height of 65 km above 80 km. We also added a second haze with a scale height of 50 km, following the analysis of de Kok et al. (2007b). This haze component (Haze A), the opacity of which is limited to the range $\sim 90\text{--}190\text{ cm}^{-1}$, is needed to reproduce a broad emission feature observed in CIRS limb spectra at various latitudes. We used the spectral dependence inferred by de Kok et al. and adjusted the integrated optical depth to best fit our two spectral selections at $5\text{--}15^\circ\text{S}$.

Fig. 7 shows the net thermal flux (upward–downward) divided by π calculated at various altitudes, with the prominent molecular bands identified. Thermal emission shortward of 600 cm^{-1} originates mostly from the troposphere, due to collision-induced absorption, but also includes a significant contribution from stratospheric haze emission. In addition, methane rotational lines are visible around 100 cm^{-1} . Beyond 600 cm^{-1} , stratospheric emission features from various rovibrational bands are observed. Continuum emission from the haze is also important. Note that, below 100 km, stratospheric emission causes a negative net thermal flux beyond 600 cm^{-1} . Fig. 8 shows the net thermal flux integrated from 10 to 1400 cm^{-1} . It can be seen that haze opacity plays an important role by increasing the radiation emitted in the stratosphere, which

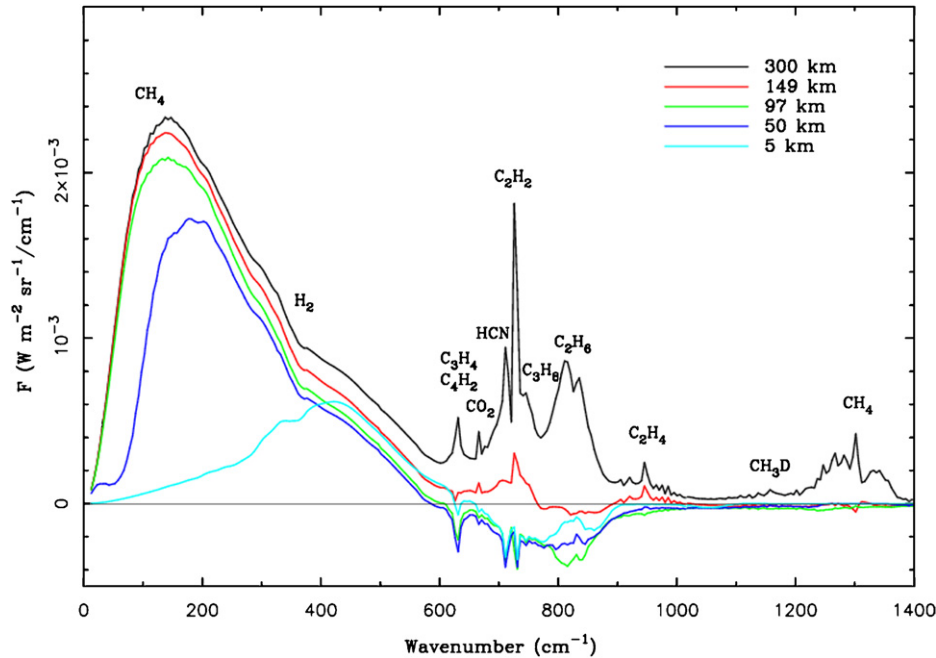


Fig. 7. $1/\pi$ times the net thermal spectral flux (πF) calculated at various altitudes in Titan's atmosphere, binned at a resolution of 5 cm^{-1} . The gaseous absorbers are indicated.

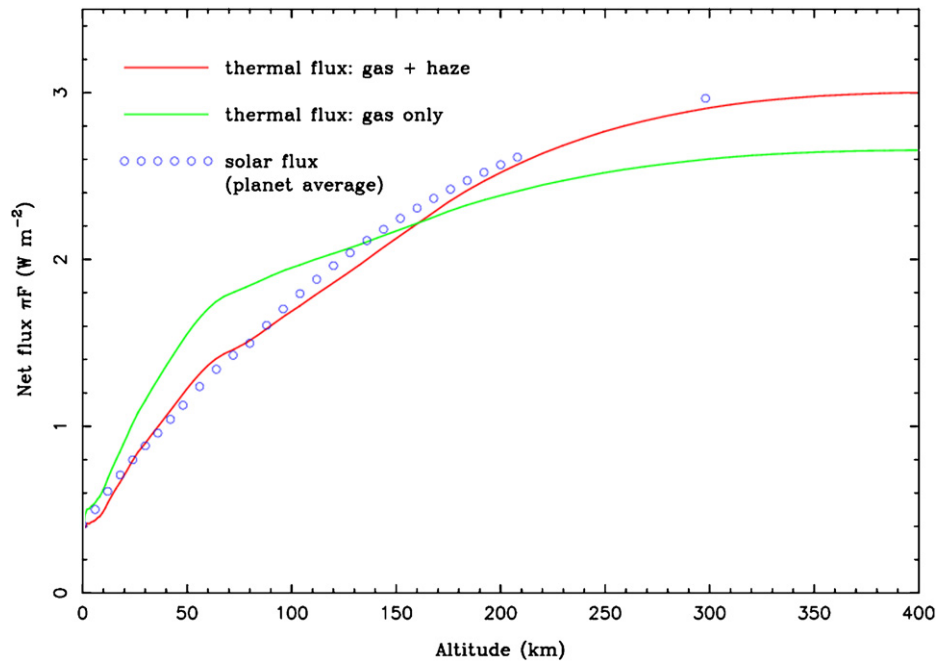


Fig. 8. Net thermal flux (πF) calculated with (red line) and without (green line) haze opacity. The net solar flux, averaged over the planet, is shown for comparison. Because the net thermal flux is defined as upward minus downward flux and the net solar flux is defined as downward flux minus upward flux, both quantities are positive in the figure, and the indicated net solar and net thermal fluxes flow in opposite directions.

adds some 15% to the planetary emission but also lowers the net thermal flux below 150 km. The sharp drop in the last couple of km near the surface is probably due to the absence of surface temperature discontinuity in the model and is therefore not reliable. Our best-fit model for the net thermal flux (positive upward) is very close to the disk-averaged net solar flux (positive downward) calculated

from DISR data. The cooling rate profile was calculated from the divergence of the net thermal flux $d(\pi F)/dp$ multiplied by mg/C_p , where m is the mean molecular mass, g the local acceleration of gravity, and C_p the specific heat of N_2 taken from the National Institute for Standards and Technology (NIST) database (<http://webbook.nist.gov>). Fig. 9 shows the so-calculated profiles. The profile that

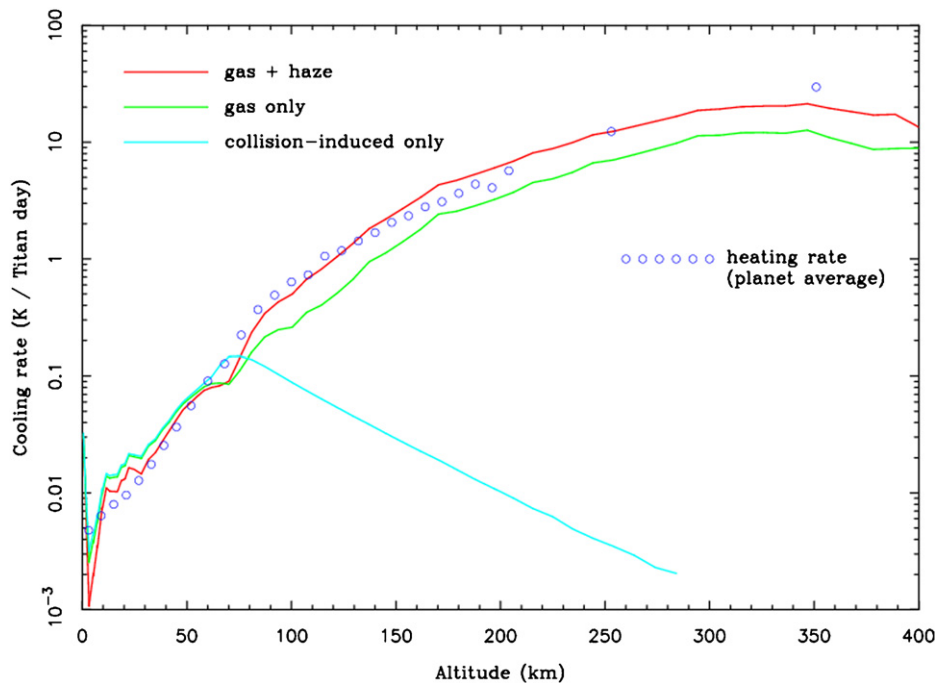


Fig. 9. Cooling rate profiles taking into account all sources of opacity (red line), only gas opacity (green line), and only collision-induced opacity (blue line). The heating rate, due to deposition of the solar flux—averaged over the planet—is shown for comparison.

incorporates all sources of opacity (gas and haze) agrees quite well with the disk-averaged heating rate profile derived from DISR measurements. While they decrease by more than 3 orders of magnitude between 350 and 0 km, both agree at all levels within a factor of 1.5 constant with altitude. Collision-induced opacity is the main source of atmospheric cooling below 80 km. Above, molecular band and haze opacities dominate, with comparable contribution from each component. Note that emission from the stratospheric haze heats the troposphere and decreases the cooling rate by about 30% below 50 km (compared to a no-haze calculation—green line in Fig. 9). The first 2 km above the surface are very sensitive to the assumed surface temperature and the sharp increase in the cooling rate from 2 to 0 km is likely due to the assumed lack of temperature discontinuity at the surface.

We think that the uncertainty on the cooling rate profile is rather small (typically 20–30%) at least up to ~400 km as our atmospheric model is constrained to reproduce the observed thermal emission at nadir and for a range of limb heights. An exception is the near-surface atmosphere (0–10 km) which is quite sensitive to the unknown surface temperature (and emissivity) and uncertainties in the $\text{CH}_4\text{--N}_2$ absorption coefficients. Another region is the 65–90 km altitude range which is not directly probed by the CIRS spectra, being intermediate between the stratospheric emission sounded by the FP3 and FP4 spectra and the upper troposphere providing most of the FP1 emission (10–600 cm^{-1}). The cooling rate in this region is sensitive to the assumed haze concentration. Changing it by a factor of 3 below 80 km around our nominal case shows a maximum

effect of 35% in the 70–80 km range and negligible influence below 65 km as collision-induced opacity then prevails. The net thermal flux profile and the cooling rate from the thermal fluxes near the latitude of the Huygens landing site are given in Table 1.

4. The net radiative heating profile at the latitude of the Huygens landing site

While the variations of the heating rate with latitude depend on the variations in haze, temperature and composition profiles with latitude, the variations at the Huygens landing latitude with longitude are thought to be small. In this case it is interesting to compare the net flux, radiative heating and cooling, and net heating profiles averaged with longitude at this location. Fig. 10 shows the solar and thermal net flux profiles averaged over a Titan day at this location. The solar net flux profile is significantly greater than the thermal net flux profile here. Fig. 11 shows the resulting solar heating and thermal cooling rates with altitude at this location. Again, the heating profile is greater than the cooling profiles at intermediate altitudes. The uncertainty in the solar heating profile is about 10% where it was directly measured (altitudes below 140 km). Finally, Fig. 12 shows the solar heating minus the thermal cooling rate profile at the latitude of the landing site averaged over one Titan day. The profile shows that heating exceeds cooling by a maximum of 0.5 K/Titan day at an altitude near 120 km and rapidly decreases above and below this altitude. Above 200 km the solar heating profile becomes uncertain as it

Table 1
Thermal net flux and radiative cooling rate

Altitude (km)	Thermal net flux (W/m ²)	Altitude (km)	Cooling rate (K/Titan day)
488.92	3.010	494.17	14.1132
478.38	3.010	483.65	12.9024
467.85	3.009	473.11	12.1524
457.31	3.009	462.58	11.0434
446.80	3.008	452.05	9.4725
436.28	3.007	441.54	9.5235
425.76	3.007	431.02	9.4469
415.25	3.006	420.51	9.1122
404.74	3.004	410.00	9.0254
394.23	3.002	399.49	13.8302
383.75	2.999	388.99	17.4412
373.25	2.995	378.50	17.2390
362.73	2.989	367.99	18.4245
352.20	2.982	357.47	19.6874
341.66	2.972	346.93	21.4900
331.13	2.961	336.39	20.5746
320.64	2.947	325.88	20.5425
310.18	2.931	315.41	20.2739
299.78	2.911	304.98	19.3542
289.45	2.888	294.62	18.8657
279.24	2.862	284.35	16.7620
269.13	2.834	274.18	15.1787
259.14	2.802	264.13	13.7493
249.25	2.767	254.19	12.4682
239.47	2.727	244.36	11.6188
229.85	2.685	234.66	9.9869
220.34	2.639	225.09	8.8690
210.96	2.587	215.65	8.1425
201.72	2.533	206.34	6.9108
192.62	2.475	197.17	6.0456
183.64	2.412	188.13	5.3430
174.83	2.344	179.24	4.7465
166.14	2.267	170.48	4.3333
157.65	2.193	161.89	3.3840
149.34	2.121	153.49	2.7216
141.24	2.049	145.29	2.2061
133.32	1.975	137.28	1.8241
125.64	1.908	129.48	1.3587
118.18	1.845	121.91	1.0480
110.93	1.784	114.56	0.8239
103.89	1.722	107.41	0.6749
97.10	1.666	100.49	0.5024
90.51	1.607	93.80	0.4292
84.18	1.549	87.34	0.3430
78.19	1.500	81.18	0.2368
72.67	1.464	75.43	0.1440
67.80	1.435	70.23	0.0906
63.70	1.404	65.75	0.0829
60.06	1.366	61.88	0.0802
56.60	1.322	58.33	0.0757
53.23	1.275	54.92	0.0670
49.90	1.224	51.56	0.0586
46.57	1.169	48.23	0.0516
43.27	1.115	44.92	0.0421
39.98	1.061	41.62	0.0341
36.67	1.007	38.32	0.0276
33.35	0.954	35.01	0.0222
29.99	0.898	31.67	0.0193
26.59	0.847	28.29	0.0146
23.12	0.777	24.85	0.0159
21.35	0.736	22.23	0.0164
19.56	0.699	20.45	0.0132

Table 1 (continued)

Altitude (km)	Thermal net flux (W/m ²)	Altitude (km)	Cooling rate (K/Titan day)
17.75	0.660	18.65	0.0127
15.90	0.625	16.82	0.0102
14.04	0.586	14.97	0.0103
12.14	0.543	13.09	0.0103
10.86	0.509	11.50	0.0111
9.56	0.482	9.56	0.0072
8.24	0.459	7.25	0.0034
7.25	0.452	5.24	0.0019
6.24	0.439	3.19	0.0011
5.24	0.430	1.60	0.0084
4.21	0.426	0.54	0.0308
3.19	0.416		
2.13	0.418		
1.07	0.387		
0.00	0.266		

depends on the vertical distribution of haze aerosols at altitudes significantly above the maximum altitude of 140 km where the Huygens probe began measurements. The solar net fluxes and the solar heating rates averaged over a Titan day are given in Table 2 at the latitude of the Huygens landing. In the troposphere, the time scales are such that averaging over a Titan year is more appropriate. Since the Huygens Probe entry was south of the equator in summer in the Southern Hemisphere, averaging over a Titan year decreases the solar heating rate slightly compared to the measured values at this latitude. Averaging over a Titan year decreases the solar heating at the Huygens landing site by about 9% from the values measured when Huygens landed, and assuming the thermal cooling rate profile is independent of solar zenith angle results in a decrease in the net radiative heating by some 30% in the low atmosphere from the values shown in Fig. 12.

The Huygens measurements were made just before the middle of summer in the Southern Hemisphere. At the 10°S latitude of the probe-landing site, the solar heating significantly exceeds the thermal cooling profile. Since there is no evidence for rapid temperature changes in this location, we assume that atmospheric dynamics acts to transport this local heating to other latitudes.

It is tempting to estimate the changes in radiative forcing for other latitudes and other seasons. For now, such estimates have to assume that cooling rates are uniform over the disk. Stratospheric temperatures are known to vary significantly with latitude as well as with the abundances of gases that are efficient infrared emitters (Flasar et al., 2005). In particular, high winter latitudes are associated with lower stratospheric temperatures (below 250 km) and larger abundances for some gases (Vinatier et al., 2007a). While it is possible that the two effects partly compensate, as inferred by Bézard et al. (1995) at the Voyager epoch, these variations question the assumption of uniform cooling rates with latitude and the radiative

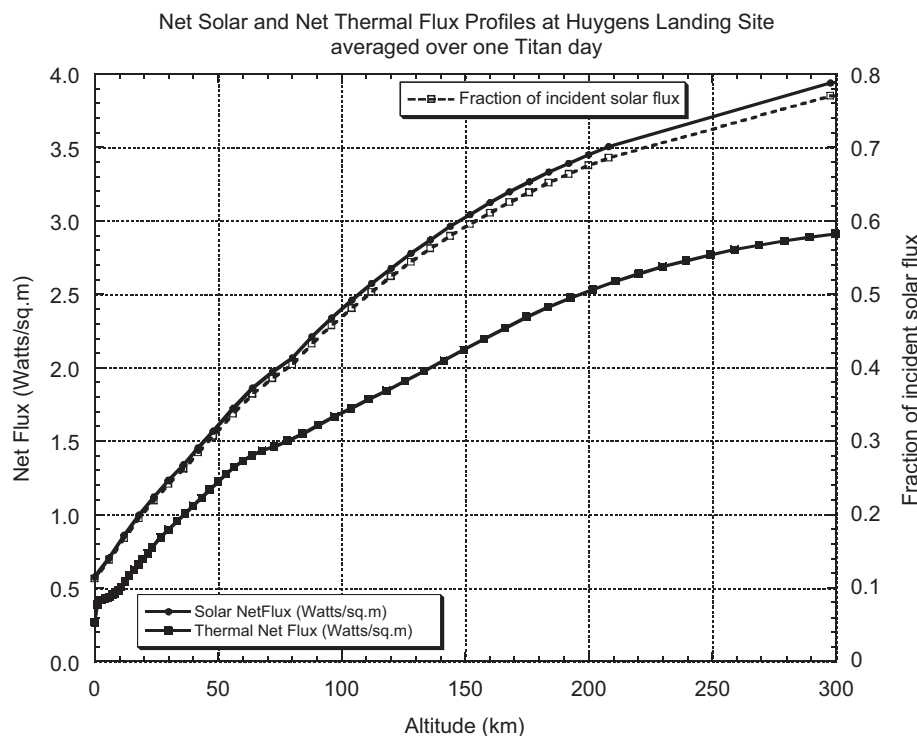


Fig. 10. The net solar flux (solid line) averaged over a Titan day is shown as a function of altitude at the 10°S latitude of the Huygens landing site. The scale on the right side of the figure gives the net flux as a fraction of the solar energy incident on this latitude of Titan (short dashed curve). The long dashed curve shows the corresponding profile of thermal net flux from CIRS measurements near the same latitude. Note that the magnitude of the solar net flux profile at this latitude significantly exceeds the magnitude of the thermal net flux profile.

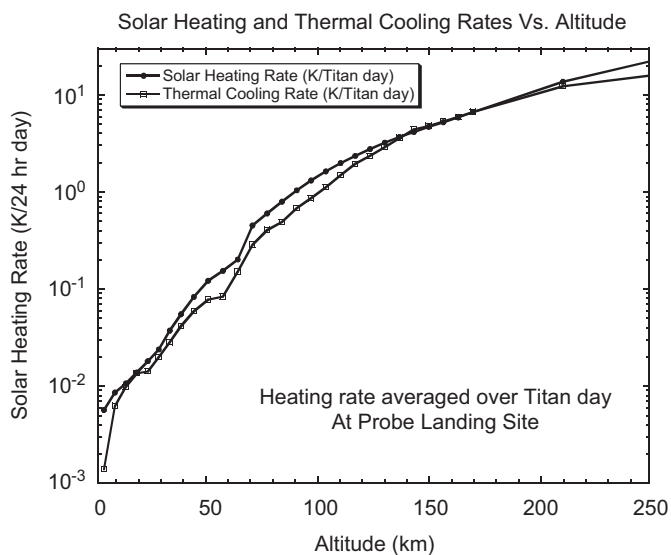


Fig. 11. The solar and thermal net flux profiles are differentiated and divided by the heat capacity to give the solar heating (curve with dots) and thermal cooling profiles (curve with squares) averaged over a Titan day at the latitude of the Huygens landing site.

forcing calculated here should be regarded as preliminary and subject to large uncertainties, especially poleward of 40°N .

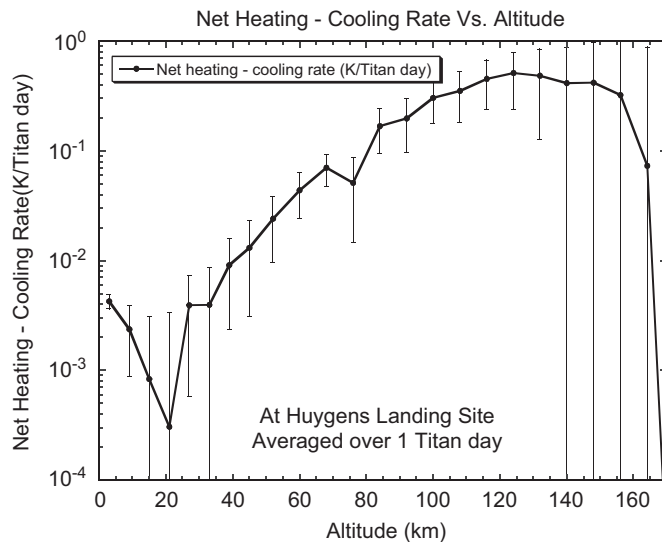


Fig. 12. The net radiative heating minus cooling profiles is shown as a function of altitude averaged over a Titan day at the latitude of the Huygens landing site. Above 170 km the solar heating profile (and the net profile) becomes less certain due to reliance on a model of haze distribution at altitudes above the 140 km altitude at which Huygens measurements began. Significant heating occurs between 70 and 160 km at this latitude at the season of the Huygens measurements. At other locations, the uncertainty in the solar heating rate is taken as 10%, while the uncertainty in the cooling rate is 20% except between 65 and 90 km and below 10 km as discussed in the text.

Table 2
Solar net flux and solar heating rate

Altitude (km)	Net solar flux (W/m ²)	Fraction of incident solar flux	Altitude (km)	Heating rate (K/Titan day)
404	4.129	0.808	351	37.3714
298	3.935	0.770	253	13.7612
208	3.504	0.686	204	6.7022
200	3.449	0.675	196	5.9172
192	3.391	0.664	188	5.2600
184	3.330	0.652	180	4.7313
176	3.265	0.639	172	4.1714
168	3.196	0.625	164	3.6833
160	3.122	0.611	156	3.2254
152	3.043	0.595	148	2.7826
144	2.960	0.579	140	2.3725
136	2.871	0.562	132	1.9899
128	2.777	0.543	124	1.6416
120	2.677	0.524	116	1.3193
112	2.571	0.503	108	1.0421
104	2.458	0.481	100	0.7999
96	2.338	0.457	92	0.6060
88	2.208	0.432	84	0.4544
80	2.067	0.404	76	0.2030
72	1.972	0.386	68	0.1547
64	1.860	0.364	60	0.1224
56	1.721	0.337	52	0.0836
48	1.569	0.307	45	0.0550
42	1.454	0.284	39	0.0375
36	1.340	0.262	33	0.0240
30	1.235	0.242	27	0.0182
24	1.120	0.219	21	0.0139
18	0.994	0.195	15	0.0107
12	0.858	0.168	9	0.0086
6	0.707	0.138	3	0.0057
0	0.574	0.112		

Averaged over longitude at probe-landing latitude.

However, making these over simplified assumptions permits crude estimates of the net radiative forcing to be made for different latitudes as functions of altitude for different seasons. For these estimates we noted the similarity of the solar heating rate averaged over the disk to the computed cooling rate at the Huygens landing latitude. Since we have no other computations of the variations of the cooling rate at other locations, we simply took the cooling rate at each location as the disk-averaged solar heating rate. Fig. 13 shows the computed solar heating rate minus the disk-averaged heating rate (taken as approximating the cooling rate) as functions of altitude for different latitudes averaged over a day at equinox. Note that the net heating rate is some 1.5 K/Titan day at 200 km altitude at the equator, and decreases to 0.035 K/Titan day near 50 km altitude. The net cooling rate at the poles is near 5 K/Titan day at 200 km altitude, decreasing to 0.06 K/Titan day near 50 km. Note the break in the heating rate profiles near 30 km due to the change in the haze structure at this altitude and the increase in the methane-mixing ratio in the troposphere.

Fig. 14 shows the net heating rate profiles near southern summer when Titan's spin axis is tipped by 25° relative to the normal to its orbital plane. Notice that at high altitudes the highest heating rate occurs at the southern pole. However, even at the pole the solar zenith angle is relatively large even at noon. At a latitude of −30°, the solar zenith angle at noon is much smaller. While the heating rate at high altitudes is not as large at −30° as at the southern pole, at lower altitudes sunlight penetrates to lower altitudes near noon and the average over a Titan rotation is greater below 50 km altitude at latitudes from −10° to −50° than at the southern pole. North of some 15° latitude thermal radiative cooling dominates solar heating.

5. Summary and discussion

We have computed the net flux profile averaged over the disk of Titan using the haze and methane absorption coefficients measured during the Huygens probe descent through the atmosphere. These measurements yielded the single-scattering phase functions of the haze particles across the solar spectrum, as well as the wavelength and altitude dependence of the single-scattering albedo and optical depth of the aerosols. The new scattering parameters are different in many ways from the models using spherical particles with constant optical properties at all altitudes and including clear regions at low altitudes in the atmosphere. Despite the changes in the detailed optical properties and distribution of the particles, the amount of solar flux absorbed at the surface (10%) is in good agreement with values used in models such as those of McKay et al. (1989) dating to times before the Cassini–Huygens mission began. The net solar flux profile averaged over the disk and the average solar heating rate is also in good agreement with these models. This agreement can serve to increase confidence in the details of the new measurements as well as in the validity of the older physical models.

The thermal cooling rate computed here near 10–15°S can also be compared to other estimates. Bézard et al. (1995) calculated the cooling rate profile in Titan's stratosphere at two latitudes, 50°N and 53°S, observed by the Voyager spacecraft. The season was then northern Spring equinox. At 53°S, in the atmospheric range mostly constrained by the thermal infrared observations, a decrease of the cooling rate from 13 K/Titan day at 0.3 mbar to 5.5 K/Titan day at 1 mbar and 2.4 K/Titan day at 3 mbar was derived. These values are close to those we infer at 15°S, respectively, 11, 5.3, and 2.0 K/Titan day. At 50°N, the cooling rates were larger than in the south, by some 40% above the 1-mbar level and some 10% at the 3-mbar level, due to the larger abundances of infrared emitters (gas and haze particles). On the other hand, the heating rate was also larger due to the enhanced haze optical depth so that radiative equilibrium could still be achieved (within the uncertainties of the model) despite the large variations in temperature and composition.

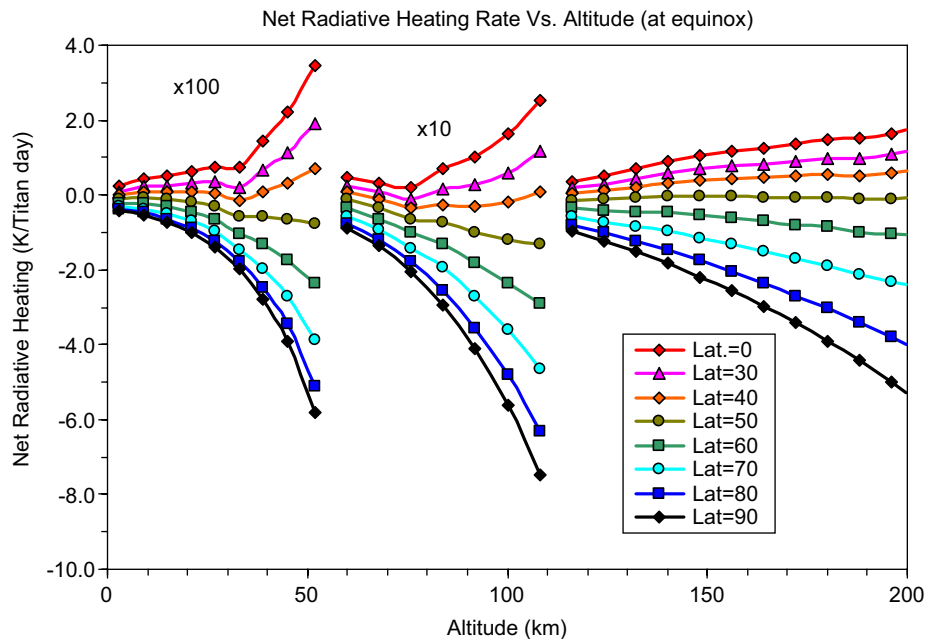


Fig. 13. A radiative cooling rate profile computed at the Huygens probe latitude is subtracted from the solar heating rate profiles averaged around latitude circles to estimate the profile of net radiative forcing at equinox. Note heating rates below 110 km are multiplied by 10 and the rates below 50 km are multiplied by 100 to make them visible on the plot. Note the break in the slopes of the curves near 30 km due to the aerosol profile and the increase in methane mixing ratio in the troposphere. These profiles are very preliminary in that they assume the haze structure and cooling rates do not vary with latitude, but they give an estimate of the magnitude of the net radiative forcing to be expected under these conditions.

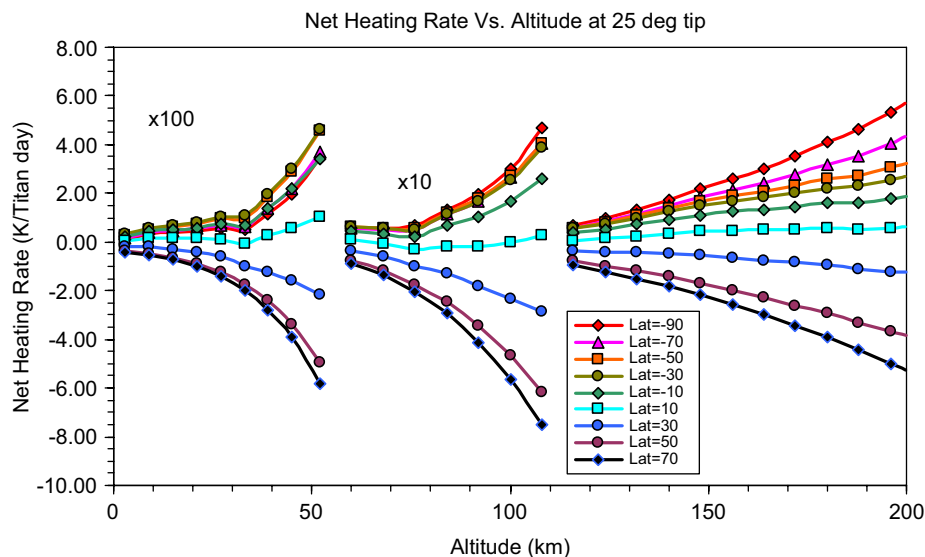


Fig. 14. Same as Fig. 13 but for the a season near southern summer solstice (Titan spin axis tipped by 25° relative to normal to plane of orbit.) Note that while maximum heating occurs at the pole at high altitudes, the maximum heating rate below 150 km occurs near a latitude of -30° due to the smaller solar zenith angle at this location at noon compared to that at the pole.

The cooling rates we derive, up to 15–20 K/Titan day in the range 250–400 km, suggest the possibility of measurable diurnal variations of temperature. At 250 km (0.25-mbar pressure level), if the zonal wind is of the order of 100 m/s, as suggested by various observations and global circulation models (Luz et al., 2006), night conditions for atmospheric parcels last about 0.06 Titan day. During the night, a radiative cooling of 0.8 K is then predicted to occur. At CIRS highest spectral resolution (0.5 cm^{-1}), the Q-branch

of the methane ν_4 band probes a region centered at 250 km (195–370 km at half-maximum of the contribution functions) for nadir-viewing observations (Vinatier et al., 2007a). A temperature change of 0.8 K at 250 km corresponds to a change of $4.0 \times 10^{-9} \text{ W/cm}^2/\text{sr/cm}^{-1}$ (i.e. 5%) of the 180-K Planck function at 1305 cm^{-1} . As this is comparable to the noise equivalent spectral radiance (NESR) of a single CIRS spectrum, such diurnal variations should be detectable through dedicated CIRS longitudinal

scans. At 150 km (2.5-mbar pressure level), still assuming a 100 m/s wind velocity, the predicted night cooling is about 0.15 K. Temperature information around this altitude can also be derived from CIRS measurements in the ν_4 band of methane even at lower spectral resolution (2.8 cm^{-1}). Temperature errors from instrument noise of about 0.2 K are then expected (Flasar et al., 2005) so that diurnal variations at this lower altitude level might also be detectable by CIRS through adequate mapping sequences at 2.8 cm^{-1} resolution.

Finally, the variation in the net radiative heating profiles at different seasons need to be compared to the values that have been used in more recent dynamical models of Titan's atmosphere. Even though it is clear that new observations by the Cassini orbiter remain to be analyzed to yield the variations in the visible haze structure as well as its thermal opacity at different latitudes and in the clearly different structures at polar latitudes, the values given here may serve as a guide in some of the basic radiative drives for these dynamical models.

Of course, these estimates of net heating rates are only the beginning of what is needed to model what is really occurring on Titan. The actual heating rate as a function of season and altitude is the result of the interaction of the atmospheric dynamics, aerosol production and microphysics, and the radiation field. The radiative time constant in Titan's lower atmosphere is much longer than a Saturn year, so the average heating rate should be closer to the seasonal average than to the daily average that we have shown in Figs. 10–14. At altitudes in the higher stratosphere, the time scales are shorter, and one might expect the temperature field to follow the daily average heating rate through the seasons. In fact, a phase lag near 90° is reported. These could be partly the result of radiative processes (Bézard et al., 1995) and of dynamical processes (Flasar and Conrath, 1990). Thus, the figures we show are simply for comparison with earlier estimates that did not include the modifying results of the interaction of atmospheric dynamics and the aerosol microphysics. In this regard, it is encouraging that they are similar to the values used in one-dimensional models. It will now be interesting to incorporate the new information on Titan's haze aerosols into new generations of three-dimensional dynamical, microphysical, and radiation models.

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